An Open-source and Optimization-based AC/DC Power Flow Model for the Brazilian Interconnected Power System

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***Abstract*—This paper introduces an open-source and optimization-based integrated AC/DC power flow model designed to analyze complex hybrid power grids, such as the Brazilian Interconnected Power System (BIPS). The BIPS is one of the biggest and most complex power grids that interconnects five regions and has high presence of Flexible AC Transmission System (FACTS) devices, High Voltage DC (HVDC) systems, specifically line-commutated converters (LCCs), and many other voltage and power flow control components. The model is formulated as a nonlinear programming (NLP) problem in which the network physics, the LCCs and FACTS devices, and the voltage and power flow controls are handled through soft constraints: the controller equations and operational limits are relaxed with bounded nonnegative slack variables, while the network power-balance equations hold as exact equalities. The model was implemented in Python using the Pyomo optimization framework and solved with the Ipopt solver, and validated against the Anarede software on two BIPS equivalent and other two large-scale BIPS power flow cases, with low and high load scenarios. Results show consistent operating points with expected numerical differences due to different assumptions and solution strategies, while demonstrating robust convergence for all the analyzed cases.**

***Index Terms*—FACTS, hybrid AC/DC systems, large-scale power systems, LCC-HVDC, LTC transformers, nonlinear programming, open-source software, phase-shifting transformers, power flow, static var compensator, switched shunts, thyristor-controlled series capacitor.**

I. Introduction

Power flow analysis is a fundamental process in power systems operations and planning, since it provides the steady-state voltages, angles, and power injections used in security assessment and grid studies [1]. For many years, this process has been solved mainly with Newton Raphson and Fast Decoupled methods, which offer good computational performance for conventional AC networks. As modern grids evolve, however, the modeling scope has expanded beyond the traditional AC setting. The growing integration of FACTS devices and HVDC systems introduces high nonlinearities, which increases the complexity of the power flow problem [2]. In hybrid AC/DC systems, representations are separated into sequential and integrated formulations. Sequential methods solve AC and DC subsystems separately and exchange boundary values until convergence [3]. This structure reduces matrix size per iteration and is attractive for large systems, but convergence can be weak in stressed or poorly conditioned cases, especially when many controls and limits are active [4].

Integrated methods solve AC and DC equations together in a single model, which improves consistency between network states and converter controls [5]. In this same approach, FACTS devices can also be embedded in the power flow equations through steady-state injection-based representations, enabling their control actions and limits to be handled within the unified network solution [6]. Discrete-type controls, such as the tap positions of on-load tap-changing (LTC) transformers and the switching of shunt banks, are commonly treated as continuous variables within the unified solution, and their discrete operating points are recovered in a post-solution step.

To increase robustness, recent studies have moved toward NLP solvers to address this issues. Compared with traditional approaches, NLP formulations are more flexible for adding limits and control equations. A common approach is the use of soft constraints with slack variables and penalty terms, which helps maintain convergence when strict feasibility is hard to satisfy in stressed operating conditions [7].

Even with these advances, there are still few studies applying this level of detailed modeling to very large systems such as the BIPS. This motivates the present work, an integrated optimization-based AC/DC power flow model with detailed LCC-HVDC, FACTS, LTC and PST representation, solved simultaneously and validated at both equivalent and full BIPS scale. The goal is to combine modeling realism with robust convergence for practical large-scale studies. The main contributions are summarized as follows:

(i) Development and open-source implementation of an integrated nonlinear AC/DC power flow model with detailed representation of network components and topology available in the original data file, solved with the Ipopt solver [8] in a single NLP framework.

(ii) Validation of the proposed model with the Anarede software, a Brazilian power system analysis tool [9], using both a system equivalent and a large-scale BIPS model, showing numerically consistent operating points and robust convergence for large-scale cases, including a high-load 2024 operating snapshot of the BIPS.

II. Problem Formulation

The AC/DC power flow model is formulated as a NLP optimization problem. While traditional root-finding methods solve a rigid system of non-linear equations, this optimization-based approach models physical system limits, such as voltage security margins and generator reactive power capabilities, directly as inequality constraints within the core network solution [10]. The algebraic structure of the problem consists of continuous decision variables bounded by technical limits, governed by a set of equality and inequality constraints that represent the unified AC and DC network physics.

A. Objective Function

In an optimization-based power flow, the objective function $f(\mathbf{x})$ can be defined as a constant to find any feasible point. In this work, the objective is defined as the squared active generation of the reference (slack) buses, whose setpoints are freed and redispatched by the solver. Because the AC power-balance equations hold as exact equalities, this criterion selects, among all feasible operating points, the one with minimum total active generation — equivalently, minimum losses for a fixed load. The physical equations of the network are therefore enforced exactly, while the controller equations (SVC droop and LTC voltage control) and the operational bounds are posed as soft constraints through bounded nonnegative slack variables, so that the model remains feasible on stressed operating points where a controller saturates at one of its limits:

$$\begin{aligned}\min_{\mathbf{x}} \quad f(\mathbf{x}) &= \sum_{g\in\mathcal{G}_s}(p_g)^2 \\ +w_{\text{ctrl}}\sum_{s\in\mathcal{S}}\Big[(r_s^+)^2+(r_s^-)^2\Big] &+ w_{\text{ctrl}}\sum_{\ell\in\mathcal{L}_{\text{ltc}}}\Big[(r_\ell^+)^2+(r_\ell^-)^2\Big] \\ +w_{\text{vlim}}\sum_{i\in\mathcal{N}}\Big[(v_i^{ub})^2+(v_i^{lb})^2\Big] &+ w_{\text{alim}}\sum_{i\in\mathcal{N}}\Big[(a_i^{ub})^2+(a_i^{lb})^2\Big] \\ +w_v\sum_{i\in\mathcal{N}}(v_i-v_i^0)^2 &+ w_a\sum_{i\in\mathcal{N}}(\theta_i-\theta_i^0)^2 \\ +w_{\text{ltc}}\sum_{\ell\in\mathcal{L}_{\text{ltc}}}(t_\ell-t_\ell^0)^2 &+ \sum_{i\in\mathcal{Q}}(qg_i-qg_i^0)^2,\end{aligned} \tag{1}$$

subject to:

$$\mathbf{g}(\mathbf{x}) = \mathbf{0}, \tag{2}$$

$$\mathbf{h}(\mathbf{x}) \le \mathbf{0}. \tag{3}$$

Where $\mathbf{x}$ is the vector of state variables, $\mathcal{G}_s$ is the set of reference (slack) generators, $\mathcal{N}$ is the set of all buses, $\mathcal{S}$ the set of SVCs, $\mathcal{L}_{\text{ltc}}$ the set of LTC transformers, and $\mathcal{Q}$ the set of reactive-power-limited generator buses. The first term minimizes the squared active generation of the slack buses, whose dispatch is determined by the exact active-power balance of the system instead of a fixed setpoint. The remaining terms are weighted penalizations of the soft-constraint slacks: (r_s^+, r_s^-) and (r_ℓ^+, r_ℓ^-) are the residual slacks of the SVC droop and LTC voltage-control equations, (v_i^{ub}, v_i^{lb}) and (a_i^{ub}, a_i^{lb}) the voltage- and angle-limit slacks, and the quadratic terms in v_i, θ_i, t_ℓ and qg_i keep the state near the input operating point $\mathbf{x}^0$. Each residual pair is bounded by its corresponding numerical tolerance, so the controller equations are recovered as exact equalities when the slacks vanish. The inequality constraints $\mathbf{h}(\mathbf{x})$ enforce the operational limits, such as tap limits and firing/extinction angles. The weights are set as $w_{\text{ctrl}} = 10^2$, $w_{\text{vlim}} = 10^2$, $w_{\text{alim}} = 10^{-2}$, $w_v = w_a = 1$, and $w_{\text{ltc}} = 10^{-2}$. The angle and voltage regularization terms are small in magnitude and only steer the solver toward the input operating point, so the converged point is not distorted when the case is already balanced.

The squared-generation dispatch criterion follows the objective proposed in [11], adapted here to a soft-constraint NLP in which the controller rows and operational bounds can release without making the exact-balance model infeasible. This objective is used consistently for all the results reported in Section IV.

B. Optimization-based AC Power Flow

In the present work, a power balance formulation in polar coordinates is adopted. The resulting steady-state equality constraints for active and reactive power balance use compact aggregated terms to represent network injections. For every bus $i \in \mathcal{N}$, where $\mathcal{N}$ represents the set of all system buses, these terms are defined as:

$$P_i^\ell := \sum_{(k,i,j)\in\mathcal{L}_i^{out}} p_{kij}^{fr} + \sum_{(k,j,i)\in\mathcal{L}_i^{in}} p_{kji}^{to}, \tag{4}$$

$$Q_i^\ell := \sum_{(k,i,j)\in\mathcal{L}_i^{out}} q_{kij}^{fr} + \sum_{(k,j,i)\in\mathcal{L}_i^{in}} q_{kji}^{to}, \tag{5}$$

$$P_i^{dc} := \sum_{h\in\mathcal{H}_i^r} P_r^{ac,h} - \sum_{h\in\mathcal{H}_i^i} P_i^{ac,h}, \tag{6}$$

$$Q_i^{dc} := \sum_{h\in\mathcal{H}_i^r} Q_r^{ac,h} + \sum_{h\in\mathcal{H}_i^i} Q_i^{ac,h}, \tag{7}$$

$$Q_i^{svc} := \sum_{s\in\mathcal{S}_i} q_s^{svc}, \tag{8}$$

$$Q_i^{sh} := \sum_{\ell\in\mathcal{S}\langle_i} q_\ell^{sh} v_i^2. \tag{9}$$

The series-compensated branches contribute to the nodal balances only through the branch-flow terms P_i^ℓ and Q_i^ℓ; the TCSC is embedded in the branch series impedance (Section II-C) and therefore introduces no separate terminal injection. Using these aggregated boundary flows, the active and reactive power balances at each bus $i \in \mathcal{N}$ are enforced as exact equalities:

$$\sum_{g\in\mathcal{G}_i} p_g - \sum_{d\in\mathcal{D}_i} p_d - g_i^{sh} v_i^2 = P_i^\ell + P_i^{dc}, \tag{10}$$

$$\sum_{g\in\mathcal{G}_i} q_g - \sum_{d\in\mathcal{D}_i} q_d + b_i^{sh} v_i^2 + Q_i^{svc} + Q_i^{sh} = Q_i^\ell + Q_i^{dc}, \tag{11}$$

The branch active and reactive flows, p_{ij} and q_{ij}, between terminal buses i and j on line $k \in \mathcal{L}$ are calculated as:

$$\begin{aligned} p_{ij} = g_{ij}&\frac{v_i^2}{t_{ij}^2} \\ &- \frac{v_i v_j}{t_{ij}}\big(g_{ij}\cos(\theta_{ij}+\phi_{ij}) + b_{ij}\sin(\theta_{ij}+\phi_{ij})\big), \end{aligned} \tag{12}$$

$$\begin{aligned} q_{ij} = -B_{ij}&\frac{v_i^2}{t_{ij}^2} \\ &+ \frac{v_i v_j}{t_{ij}}\big(g_{ij}\sin(\theta_{ij}+\phi_{ij}) - b_{ij}\cos(\theta_{ij}+\phi_{ij})\big). \end{aligned} \tag{13}$$

The transformer tap ratio t_{ij} and phase shift ϕ_{ij} in (12) and (13) enter the branch admittance of the π-model: for fixed-tap transformers and phase-shifting transformers (PST) they are fixed parameters read from the input data, while for on-load tap-changing (LTC) transformers the tap ratio is a continuous decision variable (Section II-E). Bus shunt elements are explicitly represented in the nodal balances in (10) and (11) through g_i^{sh} and b_i^{sh}, while line shunt effects are included in the branch-flow terms through $B_{ij} = b_{ij} + b_{ij}^{sh}/2$. Voltage-magnitude and angle operational limits are enforced as soft constraints:

$$v_i^{\min} - v_i^{lb} \leq v_i \leq v_i^{\max} + v_i^{ub}, \quad \forall i \in \mathcal{N}, \tag{14}$$
$$-\theta^{\lim} - a_i^{lb} \leq \theta_i \leq \theta^{\lim} + a_i^{ub}, \quad \forall i \in \mathcal{N}, \tag{15}$$

where the nonnegative slacks $v_i^{ub}, v_i^{lb}, a_i^{ub}, a_i^{lb}$ are penalized in the objective and $\theta^{\lim}$ is the reference angle limit (the default is $\theta^{\lim} = 90°$). The state variables themselves are bounded to the full feasible range of the problem, with the bus angles bounded in $(-\pi, \pi)$ so that any valid seed operating point, including the large angular separations observed across long-distance transmission corridors, can be represented. To preserve the behavior of a conventional power flow, selected state variables are fixed according to bus type by setting equal lower and upper bounds at their specified values: reference (slack) buses have both voltage magnitude and angle fixed; voltage-regulated (PV) buses have the voltage magnitude fixed, except reactive-power-limited PV buses, whose reactive generation becomes a bounded decision variable and the voltage magnitude is left free within the limits; load (PQ) buses have all state variables free.

C. Thyristor Controlled Series Capacitor (TCSC)

The impact of these devices on steady-state operation has been extensively documented, highlighting their capacity to control the power transfer on the line by acting on the terminal voltage magnitudes, the phase angles, or the line reactance [12]. In this work, the TCSC is modeled operating exclusively in constant reactance control mode. For each device $m \in \mathcal{C}$ connected between terminal buses i and j, the specified compensation defines an equivalent series reactance that is added to the series reactance x_{ij} of the compensated branch:

$$\begin{aligned} &x_m^e = x_{ij} + x_m^{csc}, \qquad x_m^{csc} = -k_m\, x_{ij}, \\ &0 \leq k_m < 1, \quad \forall m \in \mathcal{C}, \end{aligned} \tag{16}$$

where k_m is the compensation level (the ratio between the capacitive reactance provided by the device and the line reactance) and x_m^{csc} is the (negative) reactance synthesized by the device. The series impedance of the branch then becomes $r_{ij} + jx_m^e$, and the branch admittance used in the π-model of (12) and (13) is computed as:

$$g_m = \frac{r_{ij}}{r_{ij}^2 + (x_m^e)^2}, \qquad b_m = -\frac{x_m^e}{r_{ij}^2 + (x_m^e)^2}. \tag{17}$$

Because the device is purely series, it neither injects nor absorbs active or reactive power at its terminals:

$$p_m^{csc} = 0, \qquad q_m^{csc} = 0, \quad \forall m \in \mathcal{C}, \tag{18}$$

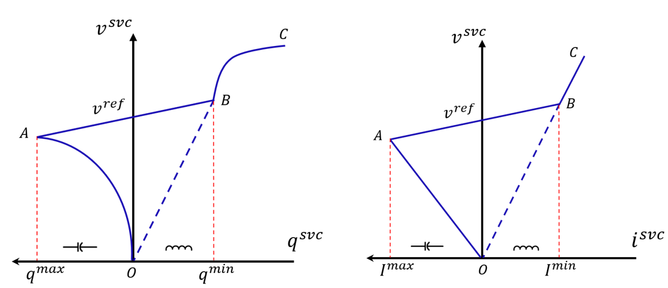
so the nodal balance equations (10) and (11) do not contain explicit CSC injection terms: the effect of the compensation is captured entirely through the modified branch flows P_i^ℓ and Q_i^ℓ. This equivalent representation keeps the NLP smooth and is consistent with how the device is reported by the input data.

D. Static VAR Compensator (SVC)

The reactive power injected or absorbed by the SVC is represented by a piecewise-linear function that captures both the control mode and the specific operating region of the device, these configurations are shown in Figure 1. Depending on whether the SVC $s \in \mathcal{S}$ is operating in reactive power control ($c_s^{svc} = 0$) or injected current control ($c_s^{svc} = 1$), different expressions are applied to characterize the interaction with the AC network, as defined in the Anarede functional specifications [12]. The reactive power injection q_s^{svc} is formulated as follows:

$$q_s^{svc} = \begin{cases} (v_s^{ref} - v_s^{svc}) v_s^t / m_s^{ctrl}, & \text{if } c_s^{svc} = 0,\ r_s^{gn} = 0 \\ (v_s^{ref} - v_s^{svc}) / m_s^{ctrl}, & \text{if } c_s^{svc} = 1,\ r_s^{gn} = 0 \\ q_s^{\min} \cdot (v_s^t)^2, & \text{if } r_s^{gn} = +1 \\ q_s^{\max} \cdot (v_s^t)^2, & \text{if } r_s^{gn} = -1 \end{cases}$$
$$\forall s \in \mathcal{S}, \tag{19}$$

where m_s^{ctrl} represents the control slope (droop) of the compensator, and r_s^{gn} is a region indicator variable identifying if the device has reached its inductive (+1) or capacitive (−1) susceptance limits. In this formulation, v_s^{ref} is the target control reference voltage for the specified bus, v_s^{svc} is the voltage at the controlled bus where the SVC is connected, v_s^t represents the terminal voltage of the SVC device, and $q_s^{\min}, q_s^{\max}$ are the minimum inductive and maximum capacitive reactive power operation limits of the compensator, respectively. In the NLP, the reactive power limit constraints are enforced as voltage-dependent bounds, $q_s^{\min} v^2 \leq q_s^{svc} \leq q_s^{\max} v^2$, and the voltage-droop equation is posed as a soft constraint whose residual slacks (r_s^+, r_s^-) are penalized in the objective. When the device saturates at one of its limits, the droop equation is released and the device keeps injecting its limiting value, reproducing the piecewise behavior above without introducing discrete variables in the model.



(a) SVC linear with Q.

(b) SVC linear with I.

Fig. 1. SVC control modes configurations.

E. Control Network Components

1) On-load Tap Changers (LTC): For each LTC transformer $\ell \in \mathcal{L}_{\text{ltc}}$ — identified in the input data by a nonzero tap range and a controlled bus — the tap ratio t_ℓ is a continuous decision variable bounded by its mechanical limits:

$$t_\ell^{\min} \le t_\ell \le t_\ell^{\max}, \quad \forall \ell \in \mathcal{L}_{\text{ltc}}. \tag{20}$$

The tap enters the branch admittance terms of (12) and (13) through $1/t_\ell^2$ on the self-side and $1/t_\ell$ on the mutual-side terms. The LTC regulates the voltage magnitude of its controlled bus toward the reference value v_ℓ^{ref}; because the tap is continuous in the NLP, the voltage target is enforced as a soft equation:

$$v_{c\ell} - v_\ell^{ref} = r_\ell^+ - r_\ell^-, \qquad r_\ell^+ + r_\ell^- \le \varepsilon_{\text{ctrl}}, \quad \forall \ell \in \mathcal{L}_{\text{ltc}}, \tag{21}$$

where $v_{c\ell}$ is the voltage magnitude of the controlled bus, $\varepsilon_{\text{ctrl}}$ is the control tolerance, and the residual slacks are penalized in the objective together with a small quadratic regularization $(t_\ell - t_\ell^0)^2$ that keeps the tap near the input setpoint. When tap optimization is disabled, t_ℓ is fixed at its input value and behaves as a parameter, which matches the fixed-tap case.

2) Phase-Shifting Transformers (PST): Phase-shifting transformers are represented by the fixed phase shift ϕ_ℓ applied through the complex tap ratio in the π-model; the trigonometric terms $\cos(\theta_{ij} + \phi_{ij})$ and $\sin(\theta_{ij} + \phi_{ij})$ in (12) and (13) directly embed the PST action in the branch flows, without additional state variables. The phase shift is read from the input data and kept fixed during the solution, which corresponds to a PST whose setpoint is defined before solving the power flow.

3) Switches and Network Reduction: Closed switches and electrically equivalent low-impedance jumper branches are contracted before the NLP is built. A switch or jumper branch is contracted when it connects buses that remain at the same voltage profile, that is, when the branch is an exact switch, or when its impedance is below the low-impedance threshold $Z_{\min}$ (default 0.001%) and the apparent flow through it exceeds 125% of its rating. The contraction merges the terminal buses with a union-find procedure: the generations, loads, and shunts of all merged buses are aggregated at the representative bus (the one with the highest bus-type priority and external degree),

$$P_i^{\text{agg}} = \sum_{j \in \mathcal{M}_i} P_j^{\text{orig}}, \qquad Q_i^{\text{agg}} = \sum_{j \in \mathcal{M}_i} Q_j^{\text{orig}}, \tag{22}$$

where $\mathcal{M}_i$ is the set of original buses merged into the representative bus i, and the contracted branches are removed from the network. A mapping back to the original buses is retained so that the solved voltages and control states are expanded to the full, uncontracted topology for reporting. This step reduces the size of the NLP without changing the physics of the solved system.

4) Switched Shunts: Bus shunt banks with automatic switching are represented by a continuous control variable q_ℓ^{sh} for each controllable bank $\ell \in \mathcal{S}\langle$, bounded by the bank limits relative to its initial injection:

$$q_\ell^{\text{sh,min}} \le q_\ell^{sh} \le q_\ell^{\text{sh,max}}, \quad \forall \ell \in \mathcal{S}\langle, \tag{23}$$

and the bank injection $q_\ell^{sh} v_i^2$ enters the reactive balance of its bus through the aggregated term Q_i^{sh} in (11). Fixed banks are treated as parameters and only contribute to b_i^{sh}. When the shunt voltage dead-band is enforced, the controlled-bus voltage magnitude is constrained by the bank's operating limits:

$$v_\ell^{\min} \le v_{c\ell} \le v_\ell^{\max}, \quad \forall \ell \in \mathcal{S}\langle, \tag{24}$$

so that the model reproduces the automatic switching behavior of the banks through the continuous NLP solution.

F. Line-Commutated Converter (LCC)

The steady-state modeling of LCC-HVDC systems used in this work follows classical converter relationships, including converter-side voltages, current, control angles, and commutation effects [13], [14]. These relationships provide the analytical basis for representing rectifier and inverter operation in AC/DC power flow studies as shown in Fig. 2. The LCC system is modeled using a unified set of equations for both rectifier ($k = r$) and inverter ($k = i$) terminals [15]. For each pole $h \in \mathcal{H}$, where $\mathcal{H}$ represents the set of all active DC poles, the DC voltage $V_k^{dc,h}$ is expressed as the ideal no-load voltage term plus the commutation reactance drop:

$$V_k^{dc,h} = N_k^h \left(\frac{3\sqrt{2}}{\pi} \frac{a_k^h V_k^{ac,h}}{T_k^h} \cos\theta_k^h + s_k \frac{3X_k^h}{\pi} I_k^{dc,h} \right), \quad \forall h \in \mathcal{H},\, k \in \{r, i\}, \tag{25}$$

where $s_r = -1$ for the rectifier and $s_i = +1$ for the inverter, which physically accounts for the polarity change and internal voltage drop across the bridges under commutation. The parameter θ_k^h denotes the operational control angle mapped to each terminal mode; specifically, $\theta_r^h = \alpha_r^h$ represents the firing delay angle for the rectifier, whereas $\theta_i^h = \gamma_i^h$ represents the minimum extinction angle for the inverter. Furthermore, N_k^h is the number of bridges that the converter station possesses for pole h at terminal k, a_k^h is the transformer ratio, T_k^h is the tap ratio, X_k^h is the commutation reactance, and $I_k^{dc,h}$ is the DC current flowing through the link.

The commutation process induces a phase displacement between the AC voltage and fundamental AC current. The corresponding overlap angle μ_k^h and power factor angle ϕ_k^h for each pole $h \in \mathcal{H}$ and terminal $k \in \{r, i\}$ are governed by:

$$\mu_k^h = \cos^{-1}\left(\cos\theta_k^h - \frac{\sqrt{2}\, I_k^{dc,h} X_k^h T_k^h}{a_k^h V_k^{ac,h}} \right) - \theta_k^h, \tag{26}$$

$$\phi_k^h = \tan^{-1}\left(\frac{\sin(2\theta_k^h + 2\mu_k^h) - \sin(2\theta_k^h) + 2\mu_k^h}{\cos(2\theta_k^h) - \cos(2\theta_k^h + 2\mu_k^h)} \right). \tag{27}$$

To guarantee mathematical and physical consistency within the NLP optimization framework, the argument of the inverse cosine in (26) is strictly constrained to the domain $[-1, 1]$, preventing non-convergence during high DC current or depressed AC voltage states.

The *RMS* AC current $I_k^{ac,h}$ at the converter transformer is proportional to the DC current, while the reactive power

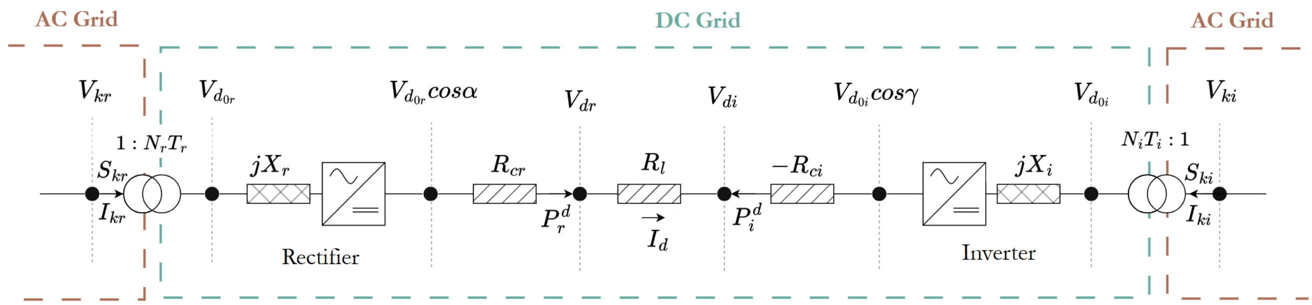


Fig. 2. LCC-HVDC converter stations and circuit representation.

variable $Q_k^{ac,h}$ at the AC interface is derived for all $h \in \mathcal{H}$ and $k \in \{r, i\}$ as follows:

$$I_k^{ac,h} = \frac{\sqrt{6}}{\pi} N_k^h I_k^{dc,h}, \tag{28}$$

$$Q_k^{ac,h} = \sqrt{3}\, I_k^{ac,h} (a_k^h V_k^{ac,h} / T_k^h) \sin \phi_k^h. \tag{29}$$

In the nodal balance, the active power terms $P_r^{ac,h}$ and $P_i^{ac,h}$ are treated as fixed setpoints specified by the system data models, representing a withdrawal at the rectifier bus and an injection at the inverter bus, respectively. Conversely, because line-commutated converters inherently consume reactive power regardless of power flow direction, both $Q_r^{ac,h}$ and $Q_i^{ac,h}$ are modeled strictly as reactive power withdrawals. These variables are directly integrated into the nodal power balance equations (10) and (11). The two-terminal DC link closes its own circuit balance: one converter is designated as the slack (folga) terminal and fixes the DC voltage of the pole, while the other terminal sets the transferred power (or current). The DC line resistance $R^{dc,h}$ allocates the losses and the terminal voltage drop:

$$P_r^{dc,h} = P_i^{dc,h} + (I^{dc,h})^2 R^{dc,h}, \tag{30}$$

$$V_r^{dc,h} = V_i^{dc,h} + I^{dc,h} R^{dc,h}. \tag{31}$$

Finally, operational feasibility and equipment limits are enforced across all components by:

$$T_k^{\min,h} \le T_k^h \le T_k^{\max,h}, \quad \forall h \in \mathcal{H},\, k \in \{r, i\}, \tag{32}$$

$$\alpha_r^{\min,h} \le \alpha_r^h \le \alpha_r^{\max,h}, \quad \forall h \in \mathcal{H}, \tag{33}$$

$$\gamma_i^{\min,h} \le \gamma_i^h \le \gamma_i^{\max,h}, \quad \forall h \in \mathcal{H}. \tag{34}$$

The converter transformer tap ratios T_k^h are continuous variables within the limits above: they adjust the AC-side voltage seen by the bridges so that the commutation angles α_r^h and γ_i^h remain small, directly minimizing the reactive power consumed by the converter stations, as further discussed in Section IV.

III. Simulation Process

The proposed model is implemented in Python using the Pyomo optimization framework and solved via the Ipopt solver. The simulation starts by parsing Anarede files (.pwf) into a readable open-source format, followed by the construction of the numerical case and the execution of the AC/DC power flow model. The complete workflow is shown in Fig. 3 and can be summarized in four stages: (i) data parsing and network reduction; (ii) Newton–Raphson warm start; (iii) NLP construction and Ipopt solve; and (iv) post-solution control restoration and reporting.

In the first stage, the .pwf file is parsed into an open-source component model, the numerical case is built, and closed switches and low-impedance jumper branches are contracted as described in Section II-E. In the second stage, a Newton–Raphson solve is attempted on the reduced case to produce a warm start for the NLP: a solved case already satisfies the balance equations and releases saturated SVC droop rows, giving Ipopt a feasible neighborhood to start from, whereas the raw Anarede snapshot is often an internally inconsistent operating point. When the Newton–Raphson iterate does not fully converge but improves the residual without diverging, its final iterate is still used as the seed, since the active rows are typically converged and only a reactive limit cycle remains. In the third stage, the NLP of Section II-B is built and solved with Ipopt [8], using the warm-started state as the initial point. In the fourth stage, the solved control variables (SVC injections, switched-shunt states, LTC taps and LCC taps and angles) are written back to the numerical case and expanded to the original, uncontracted topology for reporting.

The model reads the numerical defaults and execution options directly from the `DOPC` and `DCTE` records of the input file, reproducing the default behavior of the Anarede software. The most relevant program constants and power-flow options are summarized in Table I. The active and reactive balance tolerances of the soft constraints are derived from the Anarede interchange tolerances as $\varepsilon_p = \texttt{TEPA/BASE}$ and $\varepsilon_q = \texttt{TEPR/BASE}$ in per unit, and the control tolerance as $\varepsilon_{\text{ctrl}} = \texttt{TLVC}/100$. The convergence-related constants (`ACIT`, `VDVN`, `VDVM`, `ASTP`, `VSTP`, `CSTP`) bound the Newton–Raphson warm start and the Ipopt iteration budget, while `ZMIN` defines the low-impedance threshold used by the switch-contraction step. The activated `DOPC` options are also honored: for example, `QLIM` enables the reactive-generation limits on Q-limited PV buses, `NEWT` selects the Newton–Raphson method used in the warm start, `CTAP` enables the automatic LTC tap control, `CSCA` the automatic switching of shunt banks, `CREM` the remote voltage control, and `CPHS` the phase-shift control of PSTs.

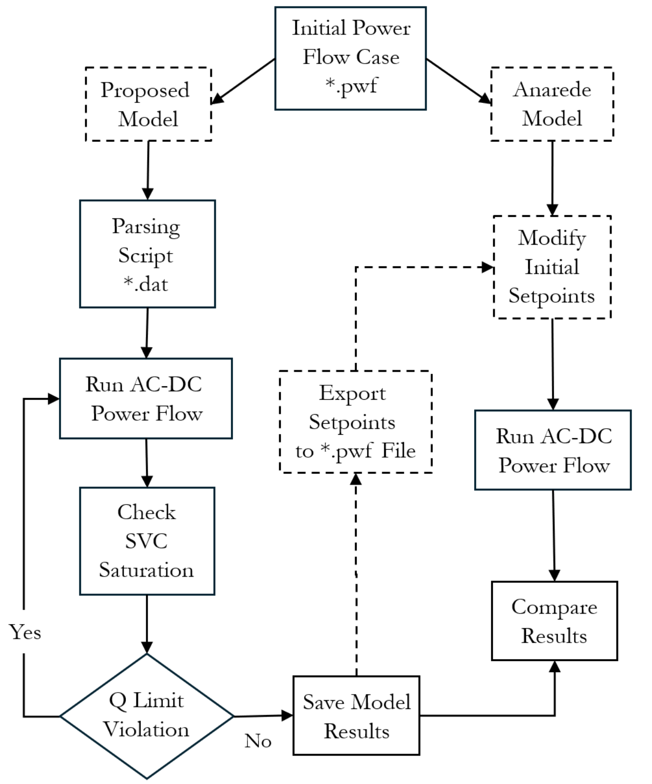


Fig. 3. Simulation workflow of the proposed optimization-based AC/DC power flow model.

TABLE I
ANAREDE PROGRAM CONSTANTS AND POWER-FLOW OPTIONS READ FROM THE `DOPC` AND `DCTE` RECORDS AND HOW THEY ARE USED BY THE PROPOSED MODEL.

Mnemonic	Default	Description	Model usage
`BASE`	100 MVA	Power base	Per-unit base of the model
`TEPA`	0.1 MW	Active int. tol.	Active balance tolerance
`TEPR`	0.1 Mvar	Reactive int. tol.	Reactive balance tolerance
`TLVC`	0.5%	Voltage ctrl. tol.	Control tolerance
`ACIT`	30	Max. iterations	Solver iteration budget
`VDVN`	40%	Min. div. voltage	NR warm-start limit
`VDVM`	200%	Max. div. voltage	NR warm-start limit
`ASTP`	0.05 rad	Max. angle step	NR step limit
`VSTP`	5%	Max. voltage step	NR step limit
`CSTP`	5%	Max. CSC step	NR step limit
`ZMIN`	0.001%	Low-impedance thr.	Switch/jumper contraction
`QLIM`	option	Q limits on PV buses	Q-limited PV variables
`CTAP`	option	LTC tap control	LTC tap optimization
`CSCA`	option	Shunt auto switching	Switched-shunt variables
`CPHS`	option	PST phase-shift ctrl.	PST setpoint read from data

IV. Numerical Results

A. BIPS Data Model

The data model used in this study is based on cases released by the Brazilian Independent System Operator [16]. Two groups of cases are analyzed: equivalent/reduced planning models and full operating snapshots. The planning models comprise `LEN_A_4_2020_SECO_2023VM_SE_EXP_N` and `LENA_BD0320R0_2Q2020_R1_CASO_12`,

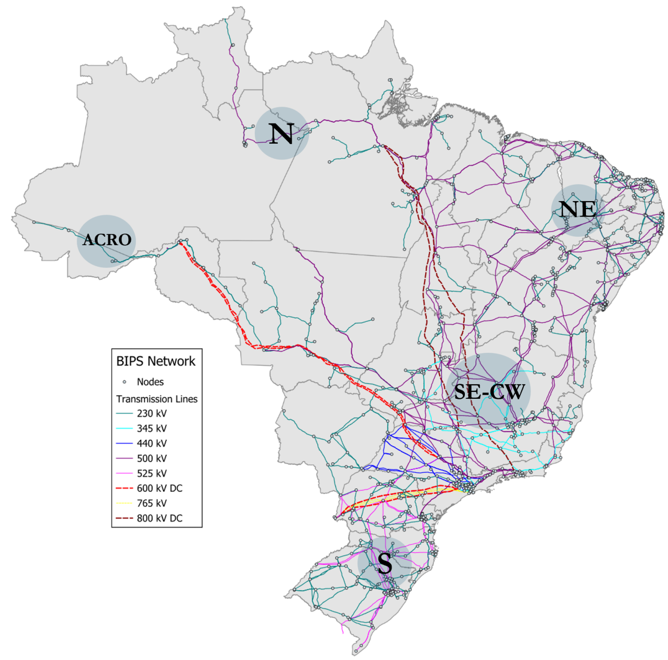


Fig. 4. BIPS transmission network.

with 7,282 and 7,291 buses, respectively, before network reduction. The operating snapshots are three 2024-08-20 cases released by the operator — `20240820_C_00-00.pwf`, `20240820_C_00-30.pwf` and `20240820_C_06-30.pwf` — each with 11,835 buses in the full parsed case (the `DBAR` block, before the closed-switch contraction), covering different loading conditions of the daily operation. The equivalent validation model `CASO_FINAL_EQV2020.pwf` has 247 buses and is used for the detailed comparison with the Anarede results. In total, the analyzed network includes the main LCC-HVDC corridors of the BIPS, together with TCSCs, SVCs, LTC transformers, PSTs, switched shunt banks, and bus and line capacitor/reactor banks.

As part of this study, the main LCC-HVDC corridors currently relevant to BIPS operation are explicitly modeled. These include the Itaipu link (Foz do Iguaçu–Ibiúna), the Madeira complex (Santo Antônio–Araraquara, ABB and Areva bipoles), and the Belo Monte transmission link (Xingu–Estreito and Xingu–Terminal Rio). In total, the implemented set comprises 12 poles organized as three major LCC schemes: four poles at 600 kV for Foz/Ibiúna, four poles at 600 kV for Santo Antônio/Araraquara, and four poles at 800 kV for the Xingu corridors, with nominal pole powers in the range of approximately 1566–2000 MW. Their representative operating settings are summarized in Table II.

B. Model Validation

To validate the proposed AC/DC power flow model, the corresponding analysis is performed on the equivalent BIPS model focusing on the point-to-point DC configuration on the LCC-HVDC links described on Table II. Using a flat start

TABLE II
LCC LINE PARAMETERS IN THE BIPS.

Line Name	**Voltage (kV)**	**Power (MW)**	**Pole No.**	α/γ (°)
Itaipu	600/587.7	1050/1018	4	15/17
Madeira 1	600/557.2	1500/1393	2	15/17
Madeira 2	600/560.4	1500/1401	2	13/18
Belo Monte 1	800/765.2	2000/1913	2	14.3/15
Belo Monte 2	800/759.5	2000/1899	2	18.4/17

configuration on the AC side, and keeping DC voltage constant on one of each converter stations, the results for the equivalent BIPS are shown in Fig. 5 and Fig. 6, respectively.

It can be observed that the proposed AC/DC power flow model tracks the AC voltage magnitudes, phase angles, generator active and reactive power outputs, branch flows, and the control setpoints of the FACTS and LCC-HVDC devices across the equivalent and full BIPS network. To quantify the similarity between the proposed AC/DC model and Anarede, an R^2 coefficient of determination was applied across the validation datasets, confirming a high fit on the results.

The mismatches observed when comparing the results with Anarede are primarily driven by differences in core modeling approaches and power flow assumptions. Because a NLP problem of this scale possesses multiple local solutions, the solver successfully converges to a physically feasible operating point around the feasible set. In addition, the warm start obtained from the Newton–Raphson step places the initial point inside the basin of attraction of the input operating point, which keeps the solution consistent with the reference case while still allowing the NLP to improve the feasibility residuals.

Furthermore, minor variations in the converter firing (α) and extinction (γ) angles are expected. While the acceptable operational limits for these angles typically span from 5° to 90°, in steady-state operation it is preferred to have small angle values for safe operation and to avoid commutation failure as shown in Table II; in this simulation, the control angles variation was in a range from 15° to 23°. Because these parameters are handled within an optimization framework, the solver's primary objective is to ensure they converge within coherent operational boundaries rather than matching the exact value. Specifically, the NLP formulation evaluates these variables from a system-wide perspective, balancing active and reactive power constraints across the entire network and optimizing the control angles to achieve a feasible solution.

Unlike the active power results, reactive power are highly sensitive to small network modeling variations. In the proposed formulation, the transformer model is simplified by applying a tap ratio configuration directly to the power flow equations. The bus and line shunts are directly added into the bus susceptance (B) values. These simplified representations can modify the localized reactive power distribution across the network. Consequently, mismatches can be observed when comparing the results with a commercial software like Anarede.

To evaluate the performance and scalability of the proposed AC/DC power flow model, the large-scale BIPS models were

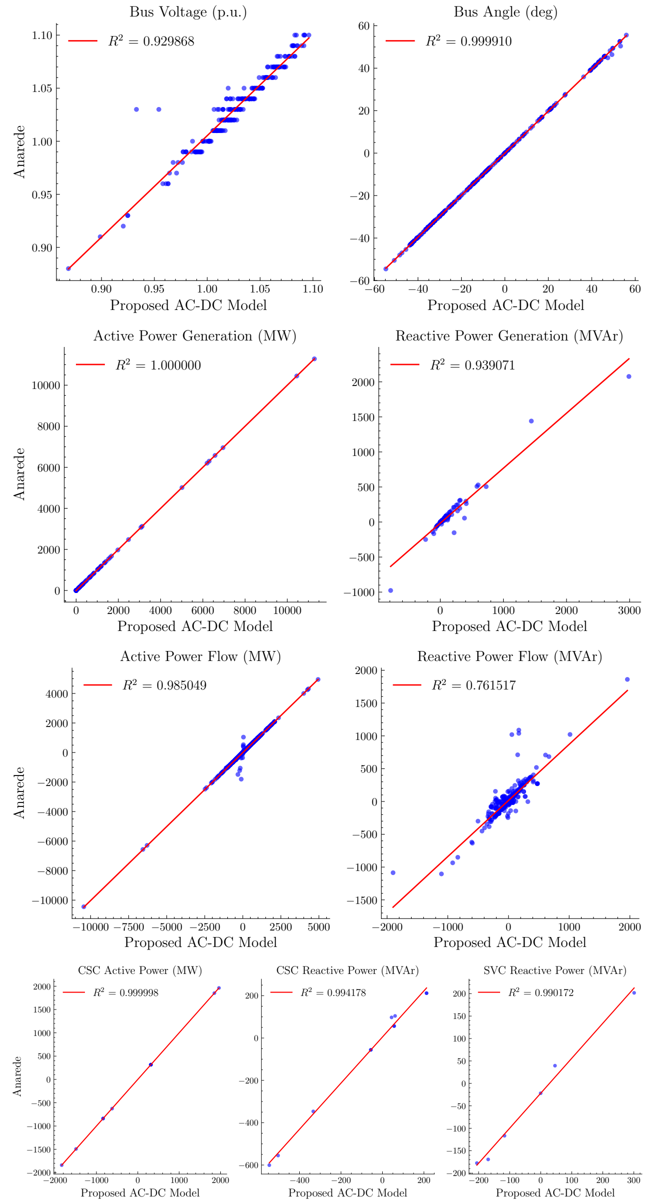


Fig. 5. Validation of equivalent BIPS model at AC components.

simulated under a normal steady-state operation mode for the LCC-HVDC. The resulting power flow profiles demonstrate that the active and reactive generator outputs, AC bus voltage magnitudes and phase angles, FACTS and LCC-HVDC terminal parameters are kept within their operational boundaries. These profiles are observed in Fig. 7 and Fig. 8.

To support the heavy transmission conditions of high active power flows transmitted over the long-distance LCC-HVDC systems, the converter transformer tap positions are controlled within an physical range of 0.9 to 1.2. Keeping the tap ratios in this range is important in normal operating mode, as they adjust the AC side voltage to minimize the converter firing angles (α) and extinction angles (γ), directly minimizing the high reactive power demand consumed by the converter stations, as can be seen in Fig. 9. Additionally, the highly meshed nature of the BIPS requires active and reactive flow

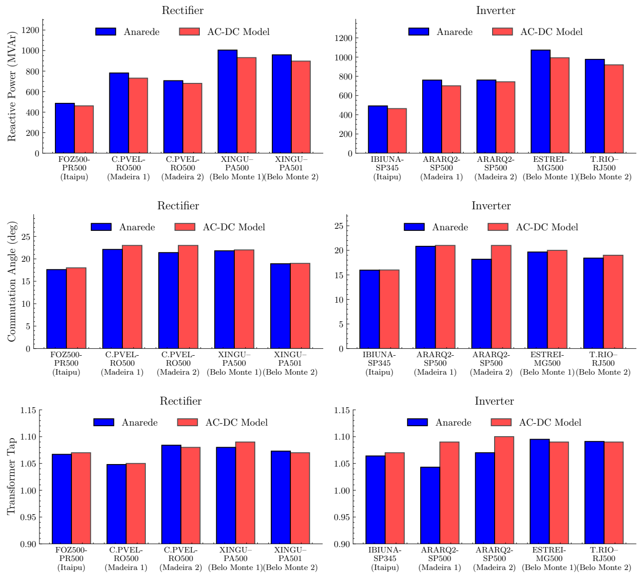


Fig. 6. Validation of equivalent BIPS model at DC components.

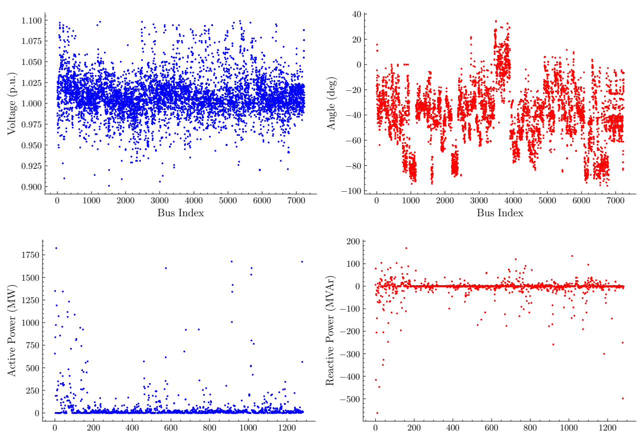


Fig. 7. AC/DC power flow results for buses and generators.

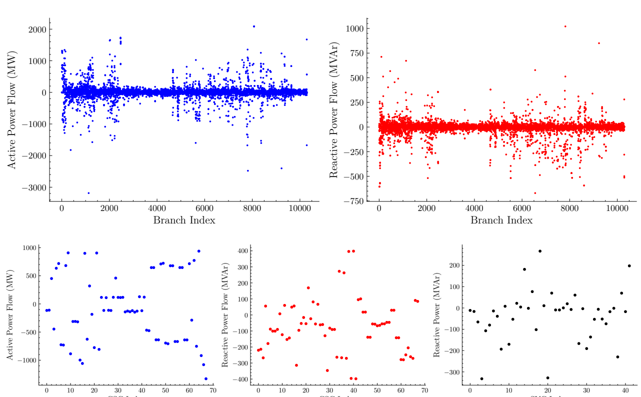


Fig. 8. AC/DC power flow results for branches and FACTS devices.

control via FACTS devices, without them the system would not converge to a feasible point. The proposed model incorporates these formulations into the nodal power balance equations. The active and reactive power through the compensated branch is controlled by the equivalent series reactance (X_{esp}), which is folded into the branch impedance as described in Section II-C. Voltage support and reactive power control are enforced at SVC buses. The model implements piecewise voltage control constraints that govern the reactive power injection (Q_c) based on localized voltage deviations ($V - V_0$) and slope characteristics (I_{ncl}), accounting for physical operating limits (Q_{cn}, Q_{cm}).

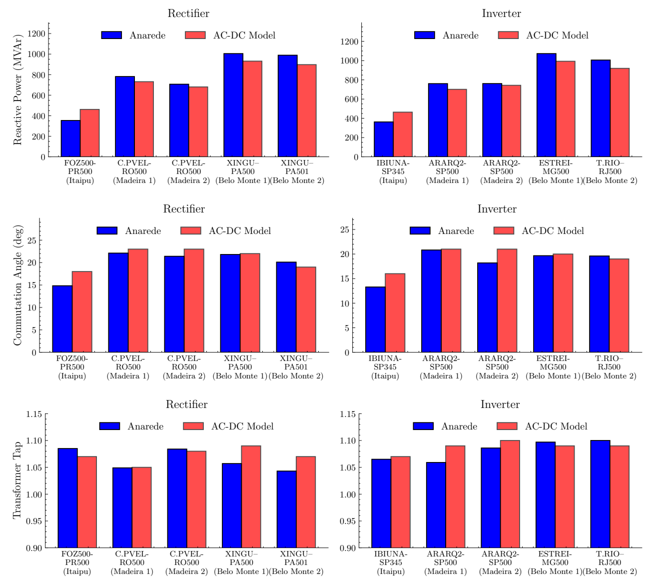


Fig. 9. Validation of BIPS model at DC components.

C. Convergence and Performance

This section reports the convergence behavior of the proposed model on all the analyzed cases, using the full `OptimizationACPowerFlow` run path. Every run follows the pipeline of Fig. 3: the closed switches are contracted, the case is warm started with the Newton–Raphson solve, and the NLP is solved with Ipopt under the squared-generation objective (Section II-B). Table III summarizes the system data of each case, reporting the number of buses in the full parsed case (the `DBAR` block, before the network reduction) and in the reduced case that the NLP is actually solved on, together with the total active and reactive generation and load (P_g, Q_g, P_l, Q_l) computed from the per-bus values of the full case.

All six runs reported in Table IV terminate with an `optimal` status of the Ipopt solver. The active and reactive power-balance residuals reach the 10^{-9}–10^{-12} band on all cases, confirming that the power-flow equations are satisfied almost to machine precision by the exact-balance formulation. The control residuals (column `max_svc`) remain at or below the control tolerance of 0.5% (`TLVC`), and the devices that saturate at their capability limits release their droop rows

TABLE III
SUMMARY OF THE ANALYZED BIPS POWER FLOW CASES.

Case	Buses	P_g (MW)	P_l (MW)	iter	time (s)
EQV-BIPS1	247	110 099	109 228	17	2
EQV-BIPS2	7 282	129 789	125 232	25	37
BIPS 00-00	11 835	77 975	74 843	58	70
BIPS 00-30	11 835	76 363	73 292	83	120

TABLE IV
CONVERGENCE RESULTS OF THE PROPOSED MODEL ON THE ANALYZED CASES UNDER THE SQUARED-GENERATION OBJECTIVE. COLUMNS: WS — NEWTON–RAPHSON WARM-START WALL TIME, IPOPT — IPOPT SOLVER TIME, TOTAL — WS+IPOPT, ITER — IPOPT ITERATIONS, AND MAX_P/MAX_Q/MAX_SVC — THE MAXIMUM ACTIVE, REACTIVE AND SVC-CONTROL RESIDUALS OF THE CONVERGED POINT.

Case	ws (s)	ipopt (s)	total (s)	iter	max_p	max_q	max_svc
LENA	29	5	34	23	1.519×10^{-9}	9.055×10^{-9}	5.463×10^{-7}
LENABD	185	13	198	28	8.895×10^{-12}	1.202×10^{-11}	1.787×10^{-8}
C 00-00	105	23	128	83	2.750×10^{-12}	6.114×10^{-12}	4.969×10^{-3}
C 00-30	927	71	998	76	2.193×10^{-11}	1.420×10^{-10}	4.600×10^{-3}
C 06-30	548	8	557	25	5.621×10^{-12}	3.228×10^{-11}	4.577×10^{-3}
CASO	< 1	< 1	< 1	17	1.044×10^{-12}	5.574×10^{-12}	2.951×10^{-3}

instead of making the model infeasible, which is accounted for in the released-control reporting.

The 2024 operating snapshots (C 00-00, C 00-30, C 06-30) require the largest computation times: their Newton–Raphson warm start is the dominant cost, ranging from roughly 10^2 s to 9×10^2 s on a single machine and a single thread, while the Ipopt solve itself completes in a few seconds up to roughly one minute. The 7,000-bus planning models converge in under one minute including the warm start, and the 247-bus equivalent solves in well under one second. The 06-30 snapshot is the daily early-morning operating point of the system and, together with the other 2024 snapshots, exercises the soft-constraint formulation on the largest network sizes analyzed, demonstrating that the proposed model converges robustly across loading conditions and system scales without retuning the weights.

V. CONCLUSIONS

This paper presents an open-source optimization-based, integrated AC/DC power flow model for the steady-state analysis of large-scale power systems incorporating LCC-HVDC and FACTS devices. The proposed model is formulated as a single, NLP problem that simultaneously solves nonlinear AC equations, DC network constraints, and converter control characteristics, together with the continuous representations of LTC transformer taps, switched shunts, phase-shifting transformers and switch topology contraction. The network power-balance equations are enforced as exact equalities, while the controller equations and operational limits are posed as soft constraints with bounded slack variables and a squared-generation objective at the reference buses selects the minimum-loss feasible point, so the solver always returns a physically meaningful operating point even under stressed conditions, while a Newton–Raphson warm start seeds the NLP from a consistent operating point and keeps the solution close to the input snapshot. The model was validated against the commercial software Anarede on a 247-bus equivalent and on large-scale models with up to 11,835 buses, including 2024 operating snapshots. All the analyzed cases converge to an optimal point of the Ipopt solver under the squared-generation objective, with active and reactive balance residuals reaching 10^{-9}–10^{-12}. While numerical differences exist between the proposed model and Anarede due to specific assumptions and component modeling, the results demonstrate that the formulation is a reliable starting point for analyzing the complexity of the BIPS.

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